3D Teeth Shape Reconstruction from a Single Face Image

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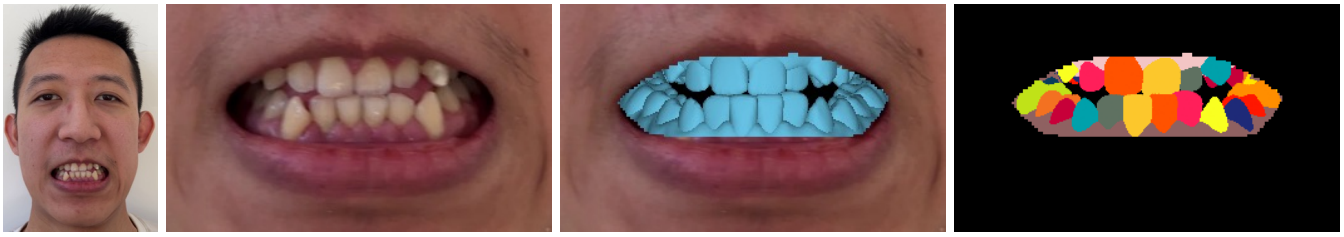


Fig. 1. **MonoTeeth.** Given a single input face image with visible teeth (the 1-st column), our method automatically localizes and extracts the tooth region to serve as the input for our neural parameter regressor (the 2-nd column). By estimating the parameters of a 3D Dental Morphable Model, we achieve high-fidelity 3D teeth shape reconstruction (the 3-rd column). Furthermore, our method naturally supports tooth-level semantic segmentation, where distinct colors represent individual tooth identities (the 4-th column).

Accurate teeth geometry is important for high-fidelity digital humans, facial animation, virtual avatars, and oral visualization. Existing teeth reconstruction methods usually require intra-oral images, multi-view capture, video sequences, or per-subject optimization, which limits their use in everyday face images. We present, to the best of our knowledge, the first feed-forward method for reconstructing personalized 3D teeth shape from a single RGB face image with visible teeth. Our method uses a pretrained vision foundation model to extract teeth-aware image features, aggregates local evidence with dental part queries, and predicts teeth shape for the upper and lower teeth. We also add an auxiliary segmentation head during training, which provides visible teeth and gum supervision and improves the use of boundary and layout cues. Because public datasets do not provide natural face images paired with accurate personalized teeth geometry, we build a synthesis pipeline that generates RGB face images, semantic masks, and known teeth shape for supervised training. Experiments show that our method produces reconstruction quality comparable to optimization-based fitting while avoiding test-time shape optimization.

CCS Concepts: • **Computing methodologies** → **Reconstruction**.

Additional Key Words and Phrases: Face Reconstruction, Teeth Reconstruction, Feed-Forward Reconstruction

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1 Introduction

High-quality 3D face modeling is a fundamental problem for digital humans, virtual reality, augmented reality, and telepresence. As the technique of facial avatars improves, recent work has increasingly moved beyond modeling the face as a single surface and instead studies specific facial components, such as hair, eyes, lips, and neck. These components have different geometries, appearances, and motion patterns and often require specialized representations. Teeth are an important but comparatively under-modeled component. Unlike skin, teeth show strong person-specific variation in shape, number, and alignment, making them a distinctive part of identity and mouth appearance. In smiles, speech, and open-mouth expressions, missing or generic teeth can quickly break the realism of a digital human. Explicit 3D teeth geometry can also support avatar editing, facial animation, dental visualization, and downstream teeth-aware performance capture. Recovering personalized teeth shape is therefore an important step toward high-quality digital humans.

Reconstructing teeth from images is more challenging than reconstructing many external facial components. In face images, teeth usually occupy only a small region and are often occluded by lips, neighboring teeth, and shadows inside the mouth. Teeth also suffer from weak surface texture, strong specular highlights, translucency, and complex intra-oral lighting. As a result, 3D teeth reconstruction from RGB appearance alone is highly under-constrained. Reliable teeth shape reconstruction typically requires additional geometric priors, boundary cues, or stronger capture conditions.

Existing methods have studied teeth reconstruction under different input settings. Methods based on intra-oral photographs can avoid lip occlusion and directly observe more complete teeth and gum structures, enabling accurate 3D reconstruction [Chen et al. 2023; Xu et al. 2024]. However, these methods require specialized devices, controlled capture procedures, and invasive acquisition, making them difficult to apply to everyday face images. Methods that rely on multi-camera capture rigs can also recover detailed inner-mouth

geometry, including teeth, gums, tongue, and jaw placement [Rasras et al. 2024]. Yet their acquisition and registration pipelines are complex and depend on expensive multi-view systems. Other methods reduce the hardware cost and reconstruct personalized teeth geometry from sparse mouth photographs or monocular phone videos [Wu et al. 2016; Zheng et al. 2025]. Nevertheless, they still rely on multiple observations and per-subject optimization.

In this paper, we present the first method for reconstructing personalized 3D teeth shape from a single RGB face image with visible teeth. Compared with intra-oral photographs, multi-view images, or video sequences, a single natural face image is the most accessible input for ordinary users and is better suited for large-scale avatar creation and everyday image applications. This setting, however, provides very limited image evidence: the teeth region is small, visibility is partial, and the single view cannot uniquely determine the complete 3D shape. Directly predicting arbitrary teeth geometry from such weak evidence is therefore unreliable. Our key insight is that the most useful image cues for teeth reconstruction are not raw appearance alone, but teeth-aware structures, such as segmentations, boundaries, and the layout of visible dental parts. Prior teeth reconstruction and inner-mouth tracking systems have shown that these cues provide effective constraints for image-based teeth geometry fitting [Rasras et al. 2024; Zheng et al. 2025]. We therefore combine such teeth-aware cues with a parametric dental model [Zhang et al. 2022], constraining the prediction to a plausible space of personalized teeth shapes.

To this end, we propose a feed-forward framework for single-image teeth shape reconstruction. We first use a vision foundation model designed for segmentation [Carion et al. 2025] to obtain teeth-aware image features from the input face image. Then, a feature decoder based on teeth-related part queries is introduced to aggregate local information for individual teeth and gums. The decoder predicts personalized teeth shapes under the constraint of a parametric dental model. During training, we additionally use auxiliary semantic segmentation supervision on visible teeth and gum regions, which provides a part-aware signal for learning spatial, boundary, and visibility information useful for 3D teeth shape prediction.

Training such a model requires supervision from paired data of both face images and personalized 3D teeth shapes. Existing public data, however, mainly focuses on oral scans, intra-oral photographs, or teeth segmentation, and does not provide paired natural face images with accurate personalized teeth geometry. To address this, we build a synthesis pipeline that generates face images with known teeth geometry and teeth semantics. This data allows us to learn the mapping from 2D face images to 3D teeth shapes in a supervised manner.

Our main contributions are:

- We present MonoTeeth, the first method for reconstructing personalized 3D teeth shape from a single RGB face image, lifting teeth geometry reconstruction toward a more accessible input setting.
- We propose a feed-forward single-image reconstruction framework that combines a pretrained vision foundation model, teeth-related part queries, and a parametric dental model to predict personalized 3D teeth shape, with auxiliary segmentation supervision used as a part-aware training signal.
- We construct a synthetic image dataset with personalized teeth shape supervision, providing training and evaluation data for single-image 3D teeth reconstruction.

2 Related Work

2.1 Face Reconstruction

Reconstructing 3D facial geometry from images is a long-standing ill-posed problem, traditionally addressed by leveraging 3D Morphable Model (3DMM) priors [Blanz and Vetter 1999; Zollhöfer et al. 2018]. Most existing methods, whether based on single-view optimization [Dib et al. 2021; Thies et al. 2016] or neural networks for 3DMM parameter regression [Deng et al. 2019; Feng et al. 2021; Wang et al. 2024], are inherently limited by the low-dimensional subspace of the 3DMM. To enhance expressiveness, several works incorporate per-vertex displacements [Lei et al. 2023; Yang et al. 2020] or fine-tune models on large-scale datasets [Han et al. 2023; Tewari et al. 2019]. Beyond parametric models, another research line explores non-parametric representations, such as UV-coordinate maps [Sela et al. 2017] or point maps [Guo et al. 2023]. While these techniques have been extended to multi-view setups to capture high-fidelity details [Han et al. 2025, 2026; Ming et al. 2026; Wu et al. 2019], they primarily focus on the external facial skin surface. Despite this significant progress, the intricate geometry of teeth remains largely neglected in these frameworks, typically being replaced by a generic proxy or ignored altogether, which limits the realism of reconstructed digital humans during oral expressions.

2.2 Teeth Reconstruction

Teeth are closely coupled with jaw motion, and several face models and capture systems therefore model the jaw as part of the facial structure. FLAME [Li et al. 2017] uses a jaw joint to articulate the lower face, while other work builds jaw rigs from multi-view cameras and markers [Yang et al. 2019; Zoss et al. 2018] or tracks jaw motion for facial performance capture [Zoss et al. 2019]. Skeleton-consistent face models further incorporate skull and jaw structure into a unified parametric representation [Qiu et al. 2022]. These methods improve the modeling of mouth articulation and facial motion, but they mainly focus on jaw pose, facial deformation, or anatomical consistency rather than reconstructing personalized teeth shape from images.

Dedicated teeth reconstruction methods use stronger dental priors or stronger capture conditions. Early works studied shape-model-based dental reconstruction from controlled images and image-based teeth tracking for orthodontic applications [Abdelrehim et al. 2013; Aichert et al. 2012]. The development of parametric dental models has made it possible to represent person-specific tooth and gum geometry in a compact space [Wu et al. 2016; Zhang et al. 2022]. Recent image-based methods reconstruct teeth from multiple intra-oral photographs [Chen et al. 2023; Wirtz et al. 2021; Xu et al. 2024], capture tooth appearance under specialized setups [Velinov et al. 2018], or track the inner mouth with multi-camera rigs [Rasras et al. 2024]. More accessible systems reduce hardware requirements and reconstruct personalized teeth geometry from sparse mouth images

or monocular phone videos [Wu et al. 2016; Zheng et al. 2025], but they still rely on multiple observations, segmentation supervision, or per-subject optimization.

In contrast, our work targets personalized 3D teeth shape reconstruction from a single RGB face image with visible teeth using a feed-forward neural network. This setting removes the need for intra-oral capture, multi-view rigs, video sequences, or test-time shape optimization, but also provides much weaker evidence because only a small and partially occluded teeth region is visible. We address this by combining a DMM prior with teeth-aware image features and synthetic paired supervision, enabling feed-forward prediction of personalized teeth shape from a natural face image.

2.3 Teeth Dataset

Public dental datasets are largely built around clinical imaging and dental scans. Two-dimensional radiograph datasets, especially panoramic X-rays, support tooth enumeration, abnormal tooth detection, and diagnosis [Hamamci et al. 2025]. Volumetric datasets based on CBCT or CT focus on tooth and maxillofacial structure segmentation with voxel-level annotations [Bolelli et al. 2026; Cui et al. 2022; Wang et al. 2025]. Intra-oral scan datasets provide 3D dental surfaces for tooth instance segmentation, FDI labeling, dental mesh understanding, and landmark detection [Ben-Hamadou et al. 2024, 2022].

These datasets have advanced dental image analysis, but they do not provide the paired supervision required by our task. Radiographs and CBCT volumes are clinical modalities rather than natural RGB face images, while intra-oral scans provide detailed tooth geometry without face appearance, lip occlusion, natural illumination, or surrounding facial context. As a result, existing public datasets cannot directly supervise the mapping from a single RGB face image to personalized 3D teeth shape. We therefore use Teeth3DS [Ben-Hamadou et al. 2022] as a source of real dental geometry and build a synthesis pipeline that produces paired RGB images, semantic masks, and DMM latent codes [Zhang et al. 2022] for supervised training.

3 Preliminaries

3.1 Dental Morphable Model

The Dental Morphable Model (DMM) [Zhang et al. 2022] is a neural SDF-based parametric model of teeth and gums. We use DMM as the shape representation for personalized teeth reconstruction. For the upper or lower teeth, DMM represents the geometry as $m = 15$ semantic components, consisting of 1 gum component and 14 tooth components excluding wisdom teeth. Each component is controlled by a latent code. Let $\mathcal{Z} = \{\mathbf{z}_i\}_{i=1}^{m}$ denote the latent code set for one row of teeth. Given a 3D query point $\mathbf{x}$, DMM predicts component-wise signed distances and semantic indicators, and combines them into an aggregated SDF:

$$F(\mathbf{x}; \mathcal{Z}) = \sum_{i=1}^{m} w_i(\mathbf{x}) F_i(\mathbf{x}; \mathbf{z}_i) \quad \text{with} \quad w_i(\mathbf{x}) = \frac{\delta_i(\mathbf{x})}{\sum_{j=1}^{m} \delta_j(\mathbf{x})}, \tag{1}$$

where F_i is the SDF of component i, δ_i is its semantic indicator, and w_i is the corresponding blending weight. The dental surface is the zero level set of the aggregated SDF:

$$\mathcal{S}(\mathcal{Z}) = \{\mathbf{x} \mid F(\mathbf{x}; \mathcal{Z}) = 0\}. \tag{2}$$

Because the model is compositional, the semantic indicators also provide part-level labels for gums and individual teeth.

In our work, we instantiate separate DMMs for the upper and lower teeth, represented by two latent code sets $\mathcal{Z}^u = \{\mathbf{z}_i^u\}_{i=1}^{15}$ and $\mathcal{Z}^l = \{\mathbf{z}_i^l\}_{i=1}^{15}$. Our network predicts these DMM shape parameters from a single image instead of directly regressing unconstrained geometry. The decoded zero-level surfaces provide personalized 3D teeth geometry, and the component indicators can be rendered as teeth and gum semantic masks.

3.2 Teeth-Inclusive Face Model

Zheng et al. [2025] reconstructs personalized teeth geometry and performs teeth-inclusive facial performance capture from a monocular phone video. Given an input video $\mathcal{I} = \{I_t\}_{t=1}^{T}$, the method estimates a face model, DMM teeth geometry, and the relative placement between the teeth and the face. Let $\boldsymbol{\phi}_t = (\boldsymbol{\beta}, \boldsymbol{\theta}_t, \boldsymbol{\psi}_t)$ denote the FLAME parameters at frame t, including identity shape, pose, and expression. The teeth are represented by DMM latent codes $\mathcal{Z}^u$ and $\mathcal{Z}^l$ for the upper and lower teeth. To assemble the teeth with the face, two rigid transformation parameters, $\mathbf{M}_{\text{up}}$ and $\mathbf{M}_{\text{low}}$, place the upper and lower DMM teeth from the canonical DMM space into the FLAME template space. We denote this teeth-inclusive parametric face model as

$$\mathcal{M}_t = \mathcal{P}_{\text{TI}}(\boldsymbol{\phi}_t, \mathcal{Z}^u, \mathcal{Z}^l, \mathbf{M}_{\text{up}}, \mathbf{M}_{\text{low}}), \tag{3}$$

where $\mathcal{P}_{\text{TI}}$ outputs a single mesh containing both the face and the teeth. In this model, the FLAME parameters control the face shape, expression, and pose, the DMM latent codes control the canonical teeth geometry, and $\mathbf{M}_{\text{up}}$ and $\mathbf{M}_{\text{low}}$ determine how the upper and lower teeth are placed in the FLAME face.

4 Method

Given a single RGB face image with visible teeth, our goal is to predict the personalized upper and lower teeth geometry. We represent the output by the DMM latent code sets $\mathcal{Z}^u$ and $\mathcal{Z}^l$ defined above. This turns single-image teeth reconstruction into a supervised prediction problem from RGB appearance to DMM shape parameters.

Existing public data do not provide face images paired with accurate personalized 3D teeth geometry. A single RGB image also provides limited evidence for the complete teeth shape, making teeth-aware structure extraction challenging. Sec. 4.1 addresses the data issue by synthesizing paired RGB images, semantic masks, and DMM parameters. Sec. 4.2 introduces a feed-forward reconstruction network for extracting teeth-aware structures and predicting DMM latent codes. Finally, Sec. 4.3 defines the training objectives.

4.1 Synthetic Training Data

Our dataset construction starts from real dental geometry. We use Teeth3DS [Ben-Hamadou et al. 2022] as a source of diverse 3D dental scans, and fit the DMM [Zhang et al. 2022] to these scans. For the n-th fitted scan, we obtain upper and lower DMM latent code sets

$$\mathcal{Z}_n = \{\mathcal{Z}_n^u, \mathcal{Z}_n^l\}, \qquad \mathcal{Z}_n^a = \{\mathbf{z}_{n,i}^a\}_{i=1}^{15}, \quad a \in \{u, l\}. \tag{4}$$

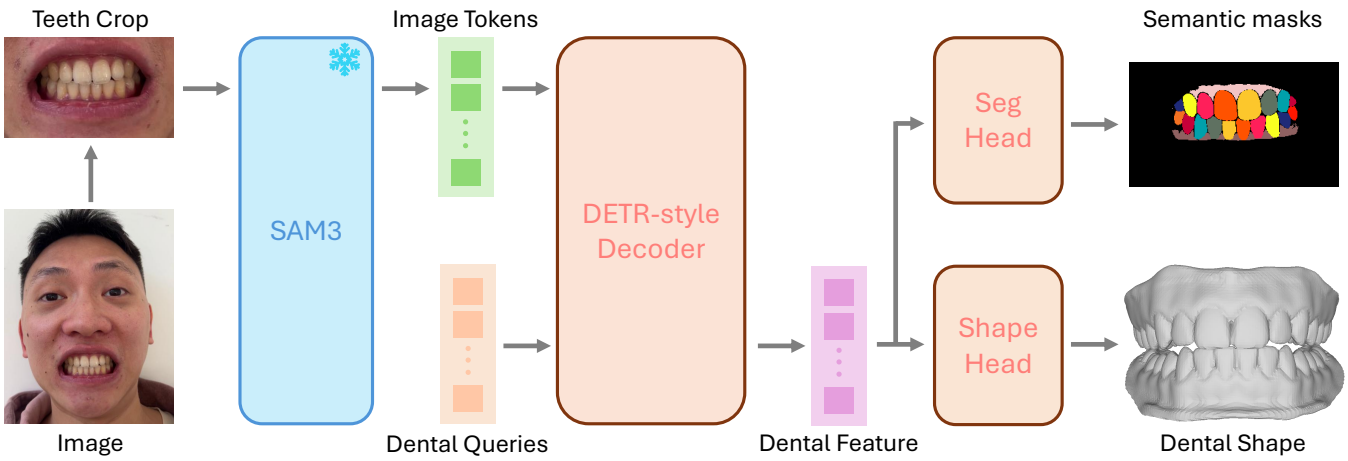


Fig. 2. **Network architecture of our method.** Given a teeth crop image automatically extracted from the captured face image, we apply a frozen SAM3 to extract image tokens. We then apply a DETR-style decoder to distill tooth-specific features into a set of dental query tokens. Next, we apply two independent heads to predict semantic masks and DMM parameters from the dental features.

These latent codes provide realistic personalized teeth geometry while keeping all targets in the same DMM parameter space.

To render the teeth in a face context, we use teeth-inclusive face templates obtained by Zheng et al. [2025]. Each template is captured from a monocular RGB video and provides a face mesh, a toothy expression, jaw and head pose, camera parameters, and the placement of the upper and lower teeth in the FLAME template space. We denote the template FLAME parameters and teeth placement by $\boldsymbol{\phi}_r$, $\mathbf{M}_{r,\text{up}}$, and $\mathbf{M}_{r,\text{low}}$. For a template r, we replace its original DMM teeth with the fitted DMM teeth $\mathcal{Z}_n^u$ and $\mathcal{Z}_n^l$. Using the notation from Eq. (3), the composed face-and-teeth mesh is

$$\widetilde{\mathcal{M}}_{r,n} = \mathcal{P}_{\text{TI}}(\boldsymbol{\phi}_r, \mathcal{Z}_n^u, \mathcal{Z}_n^l, \mathbf{M}_{r,\text{up}}, \mathbf{M}_{r,\text{low}}), \tag{5}$$

where the face parameters and teeth placement are inherited from template r, while the teeth geometry comes from scan n. This replacement gives us a face mesh with known personalized teeth geometry.

We render each composed mesh from the corresponding frontal camera view using a differentiable rasterizer [Laine et al. 2020]. The renderer produces a geometry rendering $G_{r,n}$, implemented as a surface-normal rendering in our pipeline, and semantic masks $\mathbf{Y}_{r,n}$ from the DMM component labels. We then use Nano Banana 2 [DeepMind 2026] to translate the geometry rendering into a realistic RGB face image:

$$I_{r,n} = \mathcal{G}(G_{r,n}, \mathbf{p}), \tag{6}$$

where $\mathcal{G}$ denotes the image generation model and $\mathbf{p}$ is the text prompt. The prompt asks the image generation model to preserve the facial structure specified by the geometry rendering, particularly the boundary between the teeth and gums, while adding photorealistic appearance and surrounding context. The final paired sample is

$$(I_{r,n}, \mathbf{Y}_{r,n}, \mathcal{Z}_n^u, \mathcal{Z}_n^l). \tag{7}$$

These paired data provide RGB images, semantic supervision, and DMM shape targets for training as shown in Figure 3.

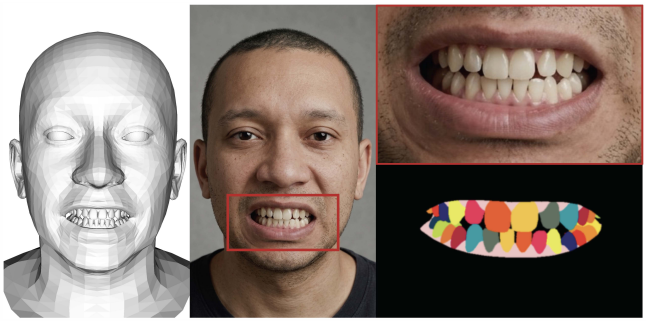


Fig. 3. **Synthetic dataset samples.** Each training pair in our dataset contains an RGB image, a semantic mask for each tooth, and the corresponding DMM parameters.

4.2 Feed-Forward Reconstruction Network

Our network predicts DMM latent codes from a single input image. Teeth are largely texture-less in face images. Therefore, raw RGB appearance alone is a weak signal for complete 3D teeth geometry. Prior teeth reconstruction methods commonly use contours, segmentations, or visible layouts as geometric constraints [Chen et al. 2023; Rasras et al. 2024; Wu et al. 2016; Zheng et al. 2025]. Following this observation, we design the network to learn teeth-aware structures useful for shape prediction.

To obtain such structures, we use a frozen SAM3 image encoder [Carion et al. 2025]. We choose SAM3 because its image encoder provides boundary-aware RGB features for separating visible teeth and gums. Given an input image I, the encoder produces image tokens,

$$\mathbf{H} = E_{\text{SAM3}}(I), \qquad \mathbf{H} \in \mathbb{R}^{N \times C}, \tag{8}$$

where N is the number of image tokens and C is the feature dimension. We keep E_{SAM3} frozen during training.

To aggregate dental-part information from the image tokens, we introduce $K = 30$ learned dental queries,

$$\mathbf{Q}_0 = \{\mathbf{q}_{0,k}\}_{k=1}^{K}, \qquad \mathbf{q}_{0,k} \in \mathbb{R}^C. \tag{9}$$

These queries correspond to the target dental parts of the upper and lower teeth, covering 1 gum component and 14 teeth for the upper teeth and the same set for the lower teeth. Following DETR [Carion et al. 2020], we associate the image tokens with positional encodings. We then feed the queries and image tokens into a DETR-style Transformer decoder:

$$\mathbf{Q} = D_\theta(\mathbf{Q}_0, \mathbf{H}), \qquad \mathbf{Q} = \{\mathbf{q}_k\}_{k=1}^{K}. \tag{10}$$

Through cross-attention, each dental query feature $\mathbf{q}_k$ collects image evidence for its corresponding dental part.

The decoder output is connected to two prediction heads. The main output branch is a DMM latent regression head. For each dental query feature $\mathbf{q}_k$, an MLP predicts a DMM latent code:

$$\widehat{\mathbf{z}}_k = h_{\text{lat}}(\mathbf{q}_k), \qquad \widehat{\mathbf{z}}_k \in \mathbb{R}^d, \tag{11}$$

where $d = 20$ is the DMM latent dimension for one component. The predicted codes are grouped into upper and lower latent code sets,

$$\widehat{\mathcal{Z}}^u = \{\widehat{\mathbf{z}}_k\}_{k=1}^{15}, \qquad \widehat{\mathcal{Z}}^l = \{\widehat{\mathbf{z}}_k\}_{k=16}^{30}. \tag{12}$$

Decoding these latent codes with DMM gives the final personalized 3D teeth geometry.

We also add a lightweight auxiliary segmentation head to guide the dental query features toward visible teeth-aware structures. Given a dental query feature $\mathbf{q}_k$ and a 2D output pixel coordinate $\mathbf{u}$, the head predicts the semantic mask logit by a dot product:

$$\widehat{Y}_k(\mathbf{u}) = h_{\text{query}}(\mathbf{q}_k)^\top h_{\text{coord}}(\gamma(\mathbf{u})), \qquad k = 1, \ldots, K. \tag{13}$$

Here $\gamma(\mathbf{u})$ is the positional encoding layer, h_{query} and h_{coord} are two MLP. Intuitively, the projected query feature represents one target dental part, while the projected coordinate feature represents one output location; their dot product gives the score of assigning that location to the dental part. This head is used only as an auxiliary training signal to encourage boundary and visible-region information in the dental queries; the final reconstruction output is the predicted DMM teeth geometry.

4.3 Training Objective

For a training sample $(I, \mathbf{Y}, \mathcal{Z}^u, \mathcal{Z}^l)$, We supervise the DMM latent codes with an ℓ_1 regression loss:

$$\mathcal{L}_{\text{lat}} = \frac{1}{K} \sum_{k=1}^{K} \left\| \widehat{\mathbf{z}}_k - \mathbf{z}_k \right\|_1. \tag{14}$$

The auxiliary segmentation branch is trained with binary cross-entropy and Dice losses:

$$\mathcal{L}_{\text{seg}} = \lambda_{\text{bce}} \mathcal{L}_{\text{BCE}}(\widehat{\mathbf{Y}}, \mathbf{Y}) + \lambda_{\text{dice}} \mathcal{L}_{\text{Dice}}(\widehat{\mathbf{Y}}, \mathbf{Y}). \tag{15}$$

The full training objective is

$$\mathcal{L} = \lambda_{\text{lat}} \mathcal{L}_{\text{lat}} + \mathcal{L}_{\text{seg}}. \tag{16}$$

In our implementation, we set $\lambda_{\text{lat}} = 10$, $\lambda_{\text{bce}} = 1$, and $\lambda_{\text{dice}} = 0.1$.

5 Experiments

5.1 Implementation Details

Datasets and preprocessing. We clean the Teeth3DS scans [Ben-Hamadou et al. 2022] by removing unsuitable samples, such as cases with wisdom teeth, resulting in 850 dental subjects. We also apply several strategies for dataset augmentation, including shape parameter interpolation and mirror flipping of teeth geometry. Ultimately, our dataset contains 2500 training pairs. For all images, we use an off-the-shelf face parsing method [Yakhyokhuja 2024] to locate the mouth region and crop a mouth-centered image as the network input. We apply the same crop to the corresponding semantic masks.

Training. We train the network with AdamW, using a learning rate of 2×10^{-4}, weight decay of 0.01, and batch size of 8. The SAM3 image encoder is kept frozen. Training takes about one day on four NVIDIA RTX 3090 GPUs.

5.2 Experimental Setup

Datasets. We evaluate our method on three types of data: synthetic images, generated images, and real images. We use the synthetic dataset described in Sec. 4.1 for training, quantitative evaluation, and qualitative evaluation. The synthetic data are split into subject-disjoint training and test sets with a 90/10 ratio. Each synthetic example contains an RGB face image, semantic masks of visible teeth and gums, and the corresponding upper and lower DMM latent codes. For out-of-distribution evaluation, we use 20 images generated by GPT-Image-2 [OpenAI 2026], 20 real images captured with phone cameras, and 20 images from Ava-256 [Martinez et al. 2024]. Because these images lack ground-truth 3D geometry, we evaluate them qualitatively and report image-space mIoU using per-tooth semantic masks obtained through SAM3-assisted manual annotation following Zheng et al. [2025]. These masks are used only for evaluation, pose alignment, and visualization; the DMM shape is predicted from RGB alone.

Baselines. We compare against an optimization-based baseline adapted from Zheng et al. [2025]. This method reconstructs teeth shape and tracks teeth motion from a monocular RGB video, with semantic masks annotated on selected frames. Since our task is single-image teeth shape reconstruction, we adapt it to a single-frame setting and provide the ground-truth semantic masks to it. We denote this baseline as *Opt. fitting*. It optimizes the DMM latent codes and teeth pose by minimizing a semantic rendering loss. We additionally evaluate *Mean-DMM fitting*, which sets all DMM latent codes to zero to fix the mean dental shape and optimizes only the upper and lower teeth poses using the same semantic rendering loss.

Metric and Evaluation. For quantitative evaluation, we measure 3D shape accuracy by the ℓ_1 Chamfer distance between reconstructed and ground-truth teeth surfaces, and image-space alignment by per-tooth semantic mIoU, both computed over visible teeth. Lower Chamfer distance and higher mIoU are better. For image-space metrics and visualization, we use the same pose-optimization procedure to project our inferred DMM shape onto the input image, keeping the DMM shape fixed.

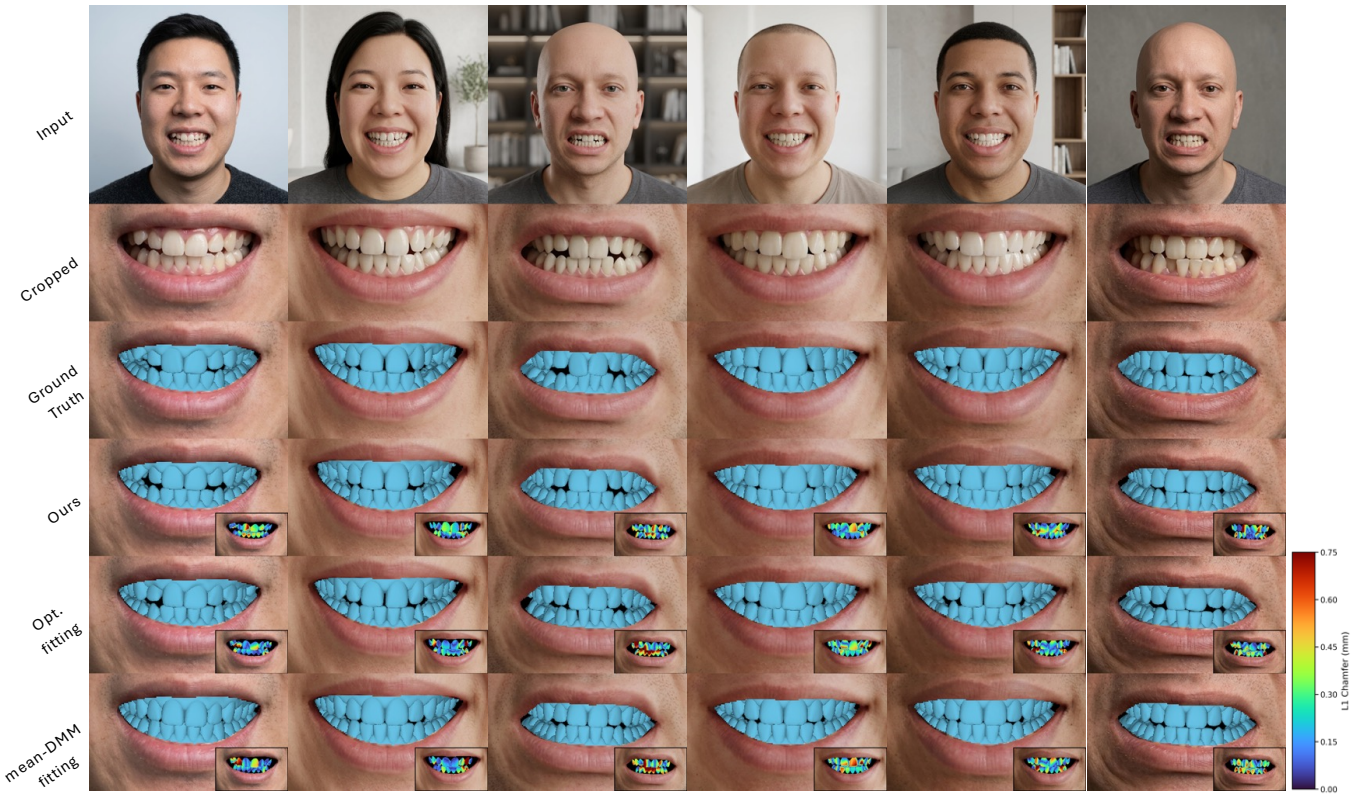


Fig. 4. Qualitative comparison on synthetic examples.

5.3 Fidelity of Synthetic Data Generation

For the synthetic supervision to remain valid, the image generation model must preserve the dental structure specified by the input geometry during photorealistic translation. To evaluate this preservation, we sample 50 images from the synthetic dataset and obtain per-tooth semantic annotations of their visible teeth through SAM3-assisted manual annotation. We compare these annotations with the corresponding semantic masks rendered from the input geometry and obtain a per-tooth semantic mIoU of 0.9167. It indicates that the image generation model largely preserves the visible tooth layout and tooth-gum boundaries during translation, supporting the use of the rendered masks and DMM parameters as paired supervision.

5.4 Results on Synthetic Images

Table 1 reports the quantitative comparison on synthetic test examples. Our feed-forward method achieves the lowest Chamfer distance among the compared settings. Compared with *Opt. fitting*, our method avoids per-image shape optimization and directly predicts the DMM latent codes from the input image. Although *Opt. fitting* achieves a higher mIoU than ours, its Chamfer distance is worse than both ours and *Mean-DMM fitting*. This result can be explained by the underdetermined nature of estimating pose and 3D shape from a single-view semantic target through differentiable rendering. A single projected mask does not uniquely constrain the teeth pose and complete 3D shape, and different pose-shape configurations can produce similar visible masks after projection. Consequently, minimizing the image-space rendering loss does not guarantee agreement with the target in 3D space. By fixing all latent codes to zero, *Mean-DMM fitting* avoids this ambiguous shape optimization and retains a plausible mean shape, yielding a lower Chamfer distance than *Opt. fitting* despite its lower mIoU.

Fig. 4 shows qualitative results on synthetic examples. Our reconstructions capture the overall size and arrangement of the visible teeth and produce plausible upper and lower teeth geometry. Compared with the optimization-based baselines, our method gives similar shape reconstructions while avoiding iterative shape fitting at test time.

5.5 Generalization to Real and Generated Images

We further evaluate generalization on real face images captured by ordinary cameras and on generated images. Because these images lack ground-truth 3D geometry, we evaluate them using per-tooth mIoU after pose alignment and qualitative comparison, as reported in Table 1 and Fig. 7, respectively. For visualization and comparison with the optimization baseline, semantic masks are used to place the predicted teeth back into the image space. The shape prediction of our method, however, is produced by a single feed-forward pass from the input RGB image.

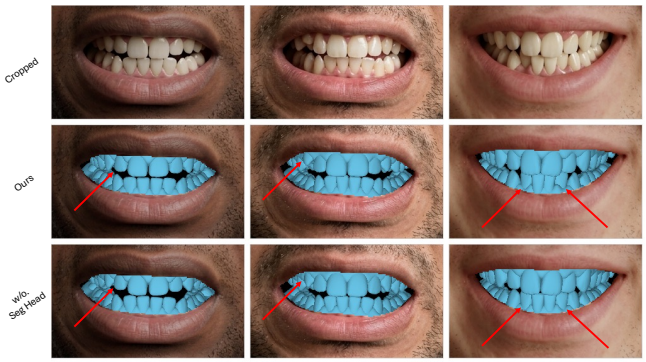


Fig. 5. Ablation study on jointly training the segmentation head.

Our method produces plausible teeth geometry for both real and generated images. These examples contain variations in identity, mouth opening, image style, and visible teeth layout, indicating that the learned teeth-aware structures transfer beyond the synthetic training images.

Table 1. Quantitative comparison and ablation studies using ℓ_1 Chamfer distance and per-tooth semantic mIoU. MAE and DINOv3 replace the SAM3 backbone, while the #subject rows vary the training-set size.

Method	Chamfer (mm) ↓	Per-tooth mIoU ↑		
		Synthetic	Generated	Real
Mean-DMM fitting	0.4879	0.7730	0.7444	0.7068
Opt. fitting	0.5382	**0.8417**	**0.8656**	**0.8366**
MAE	0.4493	0.8041	0.7205	0.7176
DINOv3	0.4425	0.8166	0.7286	0.7381
#subject 100	0.4851	0.7879	0.6991	0.7177
#subject 200	0.4557	0.8088	0.7152	0.7252
#subject 400	0.4358	0.8191	0.7538	0.7349
Ours w/o seg.	0.4346	0.8216	0.7547	0.7396
Ours	**0.4320**	0.8238	0.7598	0.7423

5.6 Ablation Study

Table 1 summarizes the ablations on the auxiliary segmentation head, image backbone, and number of training subjects. Removing the segmentation head consistently degrades both Chamfer distance and mIoU, showing that the auxiliary semantic supervision improves reconstruction performance. Using a stronger backbone and increasing the training-set size both improve the performance of our method, suggesting that the approach can scale with better vision foundation models and more training data.

As shown in Figure 5, the auxiliary supervision improves the use of visible boundary and layout cues, especially around tooth size in the first two columns and the boundary between teeth and gums in the third column. This supports our design choice of using semantic masks as an auxiliary signal rather than as the final reconstruction output.



Fig. 6. Failure cases on examples with unusual tooth shapes and arrangements. Our method produces plausible teeth geometry, but the reconstructed shape may not fully match highly distinctive dental structures.

6 Discussion

First, the reconstruction quality is bounded by the expressive power of the DMM. The parametric prior keeps the prediction in a plausible shape space, but unusual dental arrangements, missing or rotated teeth, and highly distinctive tooth shapes may not be fully represented. As shown in Fig. 6, the network may regularize such cases toward more common dental shapes, producing plausible but not fully faithful geometry. The geometry of fully invisible teeth is inferred primarily from the learned DMM prior rather than direct image evidence. Second, our feed-forward formulation does not explicitly fit the prediction to each test image. This design enables efficient reconstruction, but boundary, visibility, and pose errors cannot be corrected by test-time optimization. A promising direction is to use the predicted teeth shape and auxiliary semantic masks as initialization for lightweight image-space refinement, combining the speed of feed-forward prediction with the accuracy of optimization-based fitting when higher precision is required. The dental scans are sourced from Teeth3DS [Ben-Hamadou et al. 2022]. We will release our data-processing code and use the source datasets in accordance with their original terms.

7 Conclusion

We presented a feed-forward approach for reconstructing personalized 3D teeth shape from a single RGB face image with visible teeth. To address the lack of paired natural face images and ground-truth teeth geometry, we built a synthesis pipeline that generates RGB images, semantic masks, and DMM shape supervision from real dental scans and teeth-inclusive face templates. Our network uses a frozen vision foundation model, dental part queries, and auxiliary semantic supervision to extract teeth-aware image structures and predict upper and lower DMM latent codes. Experiments on synthetic images show that the proposed method improves over optimization-based baselines while avoiding per-image shape fitting, and qualitative results on generated and real images demonstrate plausible generalization ability beyond the synthetic training distribution. We believe this work provides a step toward accessible teeth-aware digital human modeling from ordinary face images.

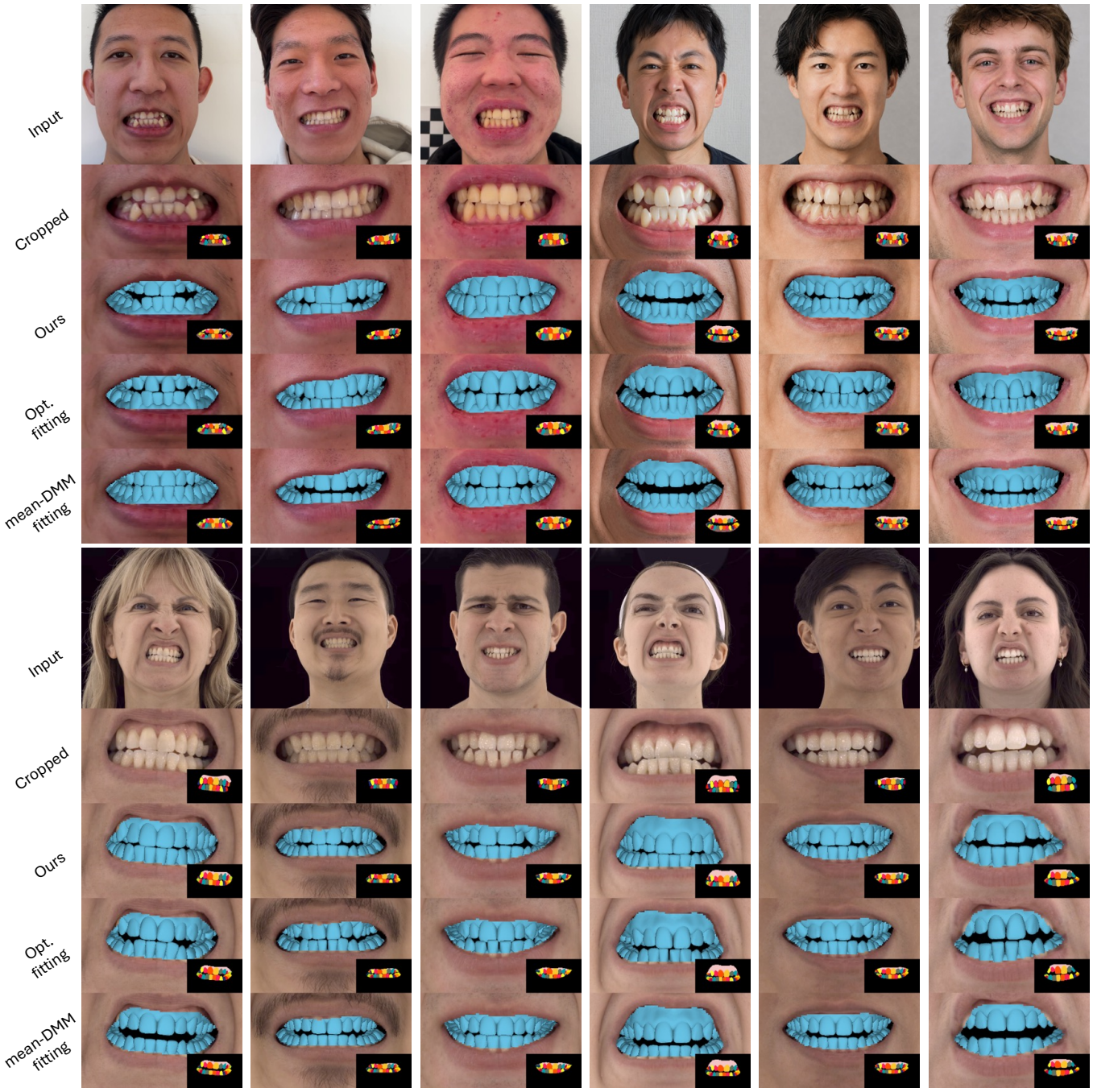


Fig. 7. Qualitative comparisons on real and generated images. In the upper group, the three subjects on the left are captured with phone cameras, and those on the right are generated by GPT-Image-2 [OpenAI 2026]; the lower group contains subjects from Ava-256 [Martinez et al. 2024]. The bottom-right insets show per-tooth semantic masks: SAM3-assisted manual annotations for the cropped inputs, and semantic renderings of the corresponding reconstructed teeth geometry for Ours and the fitting baselines.

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